Foundations of Full House Methodology

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Abstract

The Full House Methodology is an approach for computing era-adjusted or system-adjusted statistics that balances the size of the eligible population in each system with the component's performance via "versus their peers" in a specific system. We assume that each component has a latent aptitude that can be calculated from this balancing act provided the system inclusion mechanism for components of the system is known. Without making any assumptions about the distribution of the system's components, the components' distribution is well-established from the nonparametric probability distribution.

1 Introduction

Several academics have attempted to develop methods that can correct for era or system biases. The starting place for the answer to the problem of comparing players across eras can be found in the work of Stephen Jay Gould. Gould [1996] suggested that the entire distribution of achievement, or "full house of variation," is relevant when considering the achievements of baseball players across eras. This idea was also recognized as the starting place for an answer by Schell [2013] and Schell [2016].

Berry et al. [1999], Petersen et al. [2011], Schell [2013, 2016], Petersen and Penner [2020] provide several techniques to compare baseball players across eras, but they have fallen short in their objective and their results remain biased, as discussed in Eck [2020a] and Eck [2020b]. In short, these previous attempts do not incorporate the evolving talent pool in calculations, which is necessary for understanding the changing distributions of achievement over time. The result is that existing methodologies produce a disproportionate number of players who began their career before baseball was integrated players in the rankings of baseball's greatest of all time. This era of baseball did not include a crucial population of eligible players. Yu et al. [2016], Meyer [2021], Campbell [1990] also develop methods for comparing components in historical popularity, educational attainment, and evaluation of employees. But they have similar problems as stated before and fail to account for the evolving talent pool.

In some circumstances, such as test equating, it is not required to take into account an evolving talent pool. Andersson et al. [2013] offers a collaborative framework for test equating and builds the cumulative density function using the kernel method. Nevertheless,

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they are unable to resolve the extrapolation issue in non-parametric estimation, and the distribution's tail is not given special consideration.

Our Full House methodology directly incorporates the quality of the talent pool as a central modeling component. It offers metrics for assessing the performance of components in various eras or systems. With the application of educational attainment simulation, our Full House methodology reveals that the most talented students could perform worse than the other students who attend schools with lower mean scores.

The next section establishes the model setting and specifies the assumptions in our Full House methodology. Section 3 considers an intriguing paradox that the component with more talent underperforms the component with less talent. We compute the probability of this paradox under different scenarios. Section 4 discusses the application of the Full House methodology for the educational attainment simulation.

2 Model Setting

We now motivate the structure of the Full House Model. Let N_i denote the size of population i. We will suppose the every individual $k = 1, \ldots, N_i$ has an underlying latent aptitude $X_{i,1}, \ldots, X_{i,N_i} \overset{\text{iid}}{\sim} F_X$. We denote $X_{i,(j)}$ as the jth ordered latent aptitude in population i. We define $g_i(\cdot, \cdot)$ as the system inclusion function, where $g_i(X_{i,j}, \mathbf{X}_{i,-j}) = 1$ indicates that individual j is included in the ith system, and $g_i(X_{i,j}, \mathbf{X}_{i,-j}) = 0$ indicates that individual j is not included in the ith system. Let $\mathbf{X}_{i,-j}$ be the vector of all latent aptitudes not including the component j.

We now suppose that an individual j included in system i has an observed statistic $Y_{i,j}$, $j = 1, \ldots, n_i$ and we suppose that $Y_{i,j}$ is a one-to-one realization of $X_{i,j}$ given the $X_{i,j}$ is included in the ith system. Whether or not $X_{i,j}$ is chosen in the system does not alter its independence feature.

Therefore $Y_{i,1}, \ldots, Y_{i,n_i} \overset{iid}{\sim} F_{Y_i}$, F_{Y_i} is continuous, and $n_i < N_i$ is the number of individuals included in system i. We denote $Y_{i,(j)}$ as the jth ordered statistic for system i.

The key to the Full House Model is connecting latent aptitudes X with observed statistics Y. To achieve this we form pairs $(Y_{i,(j)}, X_{i,(r(N_i,n_i,j))})$, $j = 1, \ldots, n_i$, where $r(N_i, n_i, k) := r_{i,k}$ is the index function that connects this latent aptitude to the kth components in the system i. The function $r(N_i, n_i, k)$ determines the relationship between the components in the system and the selected latent aptitudes in the population. As an example, when $r(N_i, n_i, k) = N_i - n_i + k$, then the kth components in the system i is paired with the $(N_i - n_i + k)$th latent aptitude.

We now demonstrate how the Full House Model works in a parametric setting. Consider the pair $(Y_{i,(j)}, X_{i,(r(N_i,n_i,j))})$ and suppose that the distribution corresponding to $Y_{i,j}$ from the ith system is known to belong to a continuous distribution family.

In order to connect the $Y_{i,(j)}$ and $X_{i,(r(N_i,n_i,j))}$ and obtain an estimate of the underlying talents, we will make use of the following classical order statistics properties,

$$F_{Y_i}\left(Y_{i,(j)}\right) \sim U_{i,(j)}, F_{Y_{i,(j)}}\left(Y_{i,(j)}\right) \sim U_{i,j},$$

where $\sim$ means distributed as, $U_{i,j} \sim U(0,1)$, and $U_{i,(j)} \sim \text{Beta}(j, n_i + 1 - j)$.

We now connect the order statistics to the underlying talent distribution that comes from a population with $N_i \geq n_i$ observations when F_X is known. This connection is established with

$$F^{-1}_{X_{i,(r(N_i,n_i,j))}}\left(F_{Y_{i,(j)}}\left(Y_{i,(j)}\right)\right) \sim F^{-1}_{X_{i,(r(N_i,n_i,j))}}\left(U_{i,j}\right) = X_{i,(r(N_i,n_i,j))}. \tag{1}$$

Similarly, we can obtain Y from X,

$$F^{-1}_{Y_{i,(j)}}\left(F_{X_{i,(r(N_i,n_i,j))}}\left(X_{i,(r(N_i,n_i,j))}\right)\right) = Y_{i,(j)}, \tag{2}$$

3 Paradoxes Under Same Distribution

Consider two populations $i = 1, 2$ with sizes N_1 and N_2, and systems of sizes n_1 and n_2. We define a paradox (or a Gould paradox) when

$$X_{1,(r_{1,k})} > X_{2,(r_{2,j})} \quad \text{and} \quad Y_{1,(k)} < Y_{2,(j)}$$

both occur, where k, j $\in \mathbb{Z}$, $i = 1, 2$.

A central interest is evaluating the probability of such a paradox,

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right), \tag{3}$$

where $r_{i,t} = N_i - n_i + t, k = n_1, j = n_2$.

As a simple illustrative example of some mechanics concerning paradoxes, suppose that $Y_{1,j} < Y_{2,k}$ a.s. Then (3) is equivalent to $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right)$, and we have $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right) \to 1$ as $N_1 \to \infty$. We motivate a convenient representation of (3) in the following theorem.

Lemma 3.1. *Consider two populations $i = 1, 2$ with sizes N_1 and N_2, and systems of sizes n_1 and n_2. Suppose that the system inclusion mechanism has the property that $g(X_{1,(r_{1,k})}) = 1$ and $g(X_{2,(r_{2,j})}) = 1$, and that the Full House Model is of the form*

$$\begin{aligned} Y_{i,j} &\sim F_Y(y), \qquad i = 1, 2, \qquad j = 1, \ldots, n_i, \\ X_{i,j} &\sim F_X(x), \qquad i = 1, 2, \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $F_Y^{-1}(y)$ and $F_X^{-1}(x)$ exist. Then,

$$\mathrm{P}\left(X_{1,(r_{1,k})} > X_{2,(r_{2,j})}, Y_{1,(k)} < Y_{2,(j)}\right) = \mathrm{P}\left(X_{2,(r_{2,j})} < X_{1,(r_{1,k})} < h_{1,2}\left(X_{2,(r_{2,j})}\right)\right) \tag{4}$$

where $h_{1,2}(\bullet) = (f_1^{-1} \circ f_2)(\bullet)$, $f_1(\bullet) = F^{-1}_{Y_{1,(j)}}\left(F_{X_{1,(r_{1,k})}}(\bullet)\right)$, $f_2(\bullet) = F^{-1}_{Y_{2,(j)}}\left(F_{X_{2,(r_{2,j})}}(\bullet)\right)$, and both f and h exist.

Proof. If f and h exist, then we have

$$\begin{aligned} &\mathrm{P}\left(X_{1,(r_{1,k})} > X_{2,(r_{2,j})}, Y_{1,(k)} < Y_{2,(j)}\right) \\ &\quad = \mathrm{P}\left(X_{1,(r_{1,k})} > X_{2,(r_{2,j}))}, f_1(X_{1,(r_{1,k})}) < f_2(X_{2,(r_{2,j})})\right) \\ &\quad = \mathrm{P}\left(X_{2,(r_{2,j})} < X_{1,(r_{1,k})} < h_{1,2}\left(X_{2,(r_{2,j})}\right)\right). \end{aligned}$$

Existance of f_i follows directly from the existance of $F_Y^{-1}(y)$, and existance of $h_{i,j}$ follows directly from the existance of f_i and the existance of $F_X^{-1}(x)$. □

Theorem 3.2. *If we assume*

$$\begin{aligned} Y_{i,j} &\sim F_Y(y), \qquad i = 1, 2, \qquad j = 1, \ldots, n_1, \\ X_{i,j} &\sim F_X(x), \qquad i = 1, 2, \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $F_Y^{-1}(y)$ and $F_X^{-1}(x)$ are assumed to exist. If the system inclusion mechanism has the property that $g(X_{1,(r_{1,k})}) = 1$ and $g(X_{2,(r_{2,j})}) = 1$ and $g_{n_1,k} \circ g_{n_2,j}^{-1}(t) - g_{N_1,k} \circ g_{N_2,j}^{-1}(t)$ is non-negative when $t \in (0, 1)$, then

$$\begin{aligned} &\mathrm{P}\left(X_{1,(r_{1,k})} > X_{2,(r_{2,j})}, Y_{1,(k)} < Y_{2,(j)}\right) \\ &= \sum_{p=k}^{n_1} \left(\begin{array}{c} n_1 \\ p \end{array} \right) \frac{B(j+p, n_2 - n_1 - p + j + 1)}{B(j, n_2 - j + 1)} - \sum_{p=r_{1,k}}^{N_1} \left(\begin{array}{c} N_1 \\ p \end{array} \right) \frac{B(r_{2,j}+p, N_2 - N_1 - p + r_{2,j} + 1)}{B(r_{2,j}, N_2 - r_{2,j} + 1)}. \end{aligned}$$

where $B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$, $g_{n,i}(x) = I_x(i, n - i + 1)$, $I_x(\alpha, \beta)$ is the regularized incomplete beta function.

Proof.

$$\begin{aligned} &\mathrm{P}\left(X_{1,(r_{1,k})} > X_{2,(r_{2,j})}, Y_{1,(k)} < Y_{2,(j)}\right) = \mathrm{P}\left(X_{2,(r_{2,j})} < X_{1,(r_{1,k})} < h_{1,2}\left(X_{2,(r_{2,j})}\right)\right) \\ &= \int_{-\infty}^{+\infty} \left[\int_x^{F_X^{-1} \circ g_{N_1,r_{1,k}}^{-1} \circ g_{n_1,k} \circ g_{n_2,j}^{-1} \circ g_{N_2,r_{2,j}} \circ F_X(x)} f_{X_{1,(r_{1,k})}}(y)\, dy \right] f_{X_{2,(r_{2,j})}}(x)\, dx \\ &= \int_{-\infty}^{+\infty} \left[\int_x^{F_X^{-1} \circ g_{N_1,r_{1,k}}^{-1} \circ g_{n_1,k} \circ g_{n_2,j}^{-1} \circ g_{N_2,r_{2,j}} \circ F_X(x)} d\, g_{N_1,r_{1,k}}(F_X(y)) \right] f_{X_{2,(r_{2,j})}}(x)\, dx \\ &= \int_{-\infty}^{+\infty} \left[g_{n_1,k} \circ g_{n_2,j}^{-1} \circ g_{N_2,r_{2,j}} \circ F_X(x) - g_{N_1,r_{1,k}} \circ F_X(x) \right]^+ f_{X_{2,(r_{2,j})}}(x)\, dx \\ &= \int_{-\infty}^{+\infty} \left[g_{n_1,k} \circ g_{n_2,j}^{-1} \circ g_{N_2,r_{2,j}} \circ F_X(x) - g_{N_1,r_{1,k}} \circ F_X(x) \right]^+ d\, g_{N_2,r_{2,j}}(F_X(x)) \\ &= \int_0^1 \left[g_{n_1,k} \circ g_{n_2,j}^{-1}(t) - g_{N_1,r_{1,k}} \circ g_{N_2,r_{2,j}}^{-1}(t) \right]^+ dt \\ &= \int_0^1 g_{n_1,k}(x)\, dg_{n_2,j}(x) - \int_0^1 g_{N_1,r_{1,k}}(x)\, dg_{N_2,r_{2,j}}(x) \\ &= \int_0^1 I_x(k, n_1 - k + 1) \frac{x^{j-1}(1-x)^{n_2-j}}{B(j, n_2 - j + 1)}\, dx - \int_0^1 I_x(r_{1,k}, N_1 - r_{1,k} + 1) \frac{x^{r_{2,j}-1}(1-x)^{N_2-r_{2,j}}}{B(j, N_2 - j + 1)}\, dx \\ &= \int_0^1 \sum_{p=k}^{n_1} \left(\begin{array}{c} n_1 \\ p \end{array} \right) x^p (1-x)^{n_1-p} \frac{x^{j-1}(1-x)^{n_2-j}}{B(j, n_2 - j + 1)}\, dx \\ &- \int_0^1 \sum_{p=r_{1,k}}^{N_1} \left(\begin{array}{c} N_1 \\ p \end{array} \right) x^p (1-x)^{N_1-p} \frac{x^{r_{2,j}-1}(1-x)^{N_2-r_{2,j}}}{B(r_{2,j}, N_2 - r_{2,j} + 1)}\, dx \\ &= \sum_{p=k}^{n_1} \left(\begin{array}{c} n_1 \\ p \end{array} \right) \frac{B(j+p, n_2 - n_1 - p + j + 1)}{B(j, n_2 - j + 1)} - \sum_{p=r_{1,k}}^{N_1} \left(\begin{array}{c} N_1 \\ p \end{array} \right) \frac{B(r_{2,j}+p, N_2 - N_1 - p + r_{2,j} + 1)}{B(r_{2,j}, N_2 - r_{2,j} + 1)} \end{aligned}$$

where

$$\begin{aligned} h_{1,2}(x) &= (g_1 \circ f_2)(x) \\ &= F_X^{-1} \circ g_{N_1,r_{1,k}}^{-1} \circ g_{n_1,k} \circ g_{n_2,j}^{-1} \circ g_{N_2,r_{2,j}} \circ F_X(x) \\ g_{n,i}(x) &= I_x(i, n-i+1) \end{aligned}$$

$$f(x) = g_{n_1,k} \circ g_{n_2,j}^{-1}(x) - g_{N_1,k} \circ g_{N_2,j}^{-1}(x)$$

$$\begin{aligned} f^{'}(x) &= g_{n_1,k}^{'}(g_{n_2,j}^{-1}(x)) * \frac{1}{g_{n_2,j}^{'}(g_{n_2,j}^{-1}(x))} - g_{N_1,k}^{'}(g_{N_2,j}^{-1}(x)) * \frac{1}{g_{N_2,j}^{'}(g_{N_2,j}^{-1}(x))} \\ &= \frac{B(n_2-j, j+1)}{B(n_1-k, k+1)} g_{n_2,j}^{-1}(x)^{k-j}(1-g_{n_2,j}^{-1}(x))^{n_1-n_2} - \frac{B(N_2-j, j+1)}{B(N_1-k, k+1)} g_{N_2,j}^{-1}(x)^{k-j}(1-g_{N_2,j}^{-1}(x))^{N_1-N_2} \end{aligned}$$

Corollary 3.2.1. *Suppose that the system inclusion mechanism has the property that* $g(X_{1,(r_{1,k})}) = 1$ *and* $g(X_{2,(r_{2,j})}) = 1$*, and that the Full House Model is of the form*

$$\begin{aligned} Y_{i,j} &\sim F_Y(y), \qquad i = 1,2, \qquad j = 1, \ldots, n_1, \\ X_{i,j} &\sim F_X(x), \qquad i = 1,2, \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $F_Y^{-1}(y)$ *and* $F_X^{-1}(x)$ *exist. Then,*

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \begin{cases} \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2}, & \text{if } \frac{n_1}{N_1} \leq \frac{n_2}{N_2} \\ 0, & \text{otherwise} \end{cases}.$$

Corollary 3.2.2. *Suppose that the system inclusion mechanism has the property that* $g(X_{1,(r_{1,k})}) = 1$ *and* $g(X_{2,(r_{2,j})}) = 1$*, and that the Full House Model is of the form*

$$\begin{aligned} Y_{i,j} &\sim F_Y(y), \qquad i = 1,2, \qquad j = 1, \ldots, n_1, \\ X_{i,j} &\sim F_X(x), \qquad i = 1,2, \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $F_Y^{-1}(y)$ *and* $F_X^{-1}(x)$ *exist. Then,*

$$\mathrm{P}\left(X_{1,(1)} > X_{2,(1)}, Y_{1,(1)} < Y_{2,(1)}\right) = \begin{cases} \frac{N_2}{N_1+N_2} - \frac{n_2}{n_1+n_2}, & \text{if } \frac{n_1}{N_1} \geq \frac{n_2}{N_2} \\ 0, & \text{otherwise} \end{cases}.$$

Notes:

Let $n > m$. Start with

$$1 \leq \frac{g_{m,m-j}(x)}{g_{n,n-j}(x)} < +\infty$$

where the above ratio is a decreasing function in x. Now,

$$1 = \frac{x}{x} = \frac{g^{-1}_{m,m-j} \circ g_{m,m-j}(x)}{g^{-1}_{n,n-j} \circ g_{n,n-j}(x)} = \frac{g^{-1}_{m,m-j}(u_m)}{g^{-1}_{n,n-j}(u_n)}$$

where

$$u_m = g_{m,m-j}(x) \qquad u_n = g_{n,n-j}(x),$$

and we have that

$$u_m \geq u_n.$$

Thus

$$\frac{g^{-1}_{m,m-j}(u_m)}{g^{-1}_{n,n-j}(u_n)} \leq \frac{g^{-1}_{m,m-j}(u_n)}{g^{-1}_{n,n-j}(u_n)}$$

which implies that

$$g^{-1}_{n,n-j}(x) \leq g^{-1}_{m,m-j}(x)$$

This implies that

$$x \leq g_{n,n-j} \circ g^{-1}_{m,m-j}(x)$$

and

$$g_{m,m-j} \circ g^{-1}_{n,n-j}(x) \leq x$$

Lemma 3.3. *Consider two populations $i = 1, 2$ with sizes N_1 and N_2, and systems of sizes n_1 and n_2. Suppose that the system inclusion mechanism has the property that $g(X_{i,j}, \boldsymbol{X}_{i,-j}) = I(X_{i,j} \geq X_{N_i - n_i + 1})$ for $i = 1, 2$, and that the Full House Model is of the form*

$$\begin{aligned} Y_{i,j} &\sim F_Y(y), \qquad i = 1, 2, \qquad j = 1, \ldots, n_i, \\ X_{i,j} &\sim F_X(x), \qquad i = 1, 2, \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $F^{-1}_Y(y)$ and $F^{-1}_X(x)$ exist. Suppose that $n_2 \geq n_1 > 1$, $N_1 \geq N_2 > 1$, and $k = j$. Then $g_{n_1,k} \circ g^{-1}_{n_2,j}(t) - g_{N_1,k} \circ g^{-1}_{N_2,j}(t)$ is non-negative when $t \in (0, 1)$.

Proof. This proof will be by two induction arguments. For the base case, let $n_1 = n_2$ and $N_1 = N_2$. With these specifications we have that

$$g_{n_1,k} \circ g^{-1}_{n_2,j}(t) - g_{N_1,k} \circ g^{-1}_{N_2,j}(t) = 0.$$

For the induction step, choose some $m \leq n_1 - 1$, and suppose that $g_{n_1-m,k} \circ g^{-1}_{n_2,j}(t) - g_{N_1,k} \circ g^{-1}_{N_2,j}(t)$ is non-negative for $t \in (0, 1)$ where $n_2 \geq n_1$ and $N_1 \geq N_2$. Since $g_{n,k}$ is the cumulative distribution function for a Beta$(n - k, k + 1)$ random variable we can write

$$g_{n_1-m-1,k} \circ g^{-1}_{n_2,j}(t) - g_{N_1,k} \circ g^{-1}_{N_2,j}(t)$$

$$= g_{n_1-m,k} \circ g_{n_2,j}^{-1}(t) - g_{N_1,k} \circ g_{N_2,j}^{-1}(t) + \frac{\left[g_{n_2,j}^{-1}(t)\right]^{n_1-1}\left[1-g_{n_2,j}^{-1}(t)\right]^{k+1}}{(n_1-1)B(n_1-1,k+1)}$$

$$\geq 0,$$

where the right hand sign of the equality is a property of regularized incomplete beta functions https://dlmf.nist.gov/8.17. Thus the induction step is satisfied. A similar argument holds for iterative increases to N_1. □

Proposition 3.4. *Suppose* $\frac{n_1}{N_1} \geq 1 \geq \frac{n_2}{N_2}$ *and* $k = j$*, then* $g_{n_1,k} \circ g^{-1}_{n_2,j}(t) - g_{N_1,k} \circ g^{-1}_{N_2,j}(t)$ *is non-negative when* $t \in (0,1)$*.*

Proof. Define the function $f(x) = g_{n_1,k}(x) - g_{N_1,k}(x), x \in (0,1)$ and we find $f(0) = 0$ and $f(1) = 0$. Then when $n_1 \geq N_1$

$$\begin{aligned} f^{`}(x) &= \frac{x^{k-1}(1-x)^{n_1-k}}{B(k, n_1+1-k)} - \frac{x^{k-1}(1-x)^{N_1-k}}{B(k, N_1+1-k)} \\ &= \frac{x^{k-1}(1-x)^{N_1-k}}{B(k, n_1+1-k)}((1-x)^{n_1-N_1} - \frac{B(k, n_1+1-k)}{B(k, N_1+1-k)}) \end{aligned}$$

When $x = 0$, $(1-x)^{n_1-N_1} = 1$ and when $x = 1$, $(1-x)^{n_1-N_1} = 0$. Also $\frac{B(k,n_1+1-k)}{B(k,N_1+1-k)} \leq 1$.

Then $f^{`}(x)$ is positive first and then becomes negative. And now we have $f(x) \geq 0$ for all $x \in (0,1)$.

Then when $x = g^{-1}_{n_2,j}(t)$, where $t \in (0,1)$, we have

$$g_{n_1,k}(g^{-1}_{n_2,j}(t)) \geq g_{N_1,k}(g^{-1}_{n_2,j}(t)) \tag{5}$$

Since $g_{N_1,k}(x)$ is an increasing function and $g^{-1}_{n_2,j}(t) \geq g^{-1}_{N_2,j}(t)$ when $n_2 \leq N_2$

Now we have

$$g_{N_1,k}(g^{-1}_{n_2,j}(t)) \geq g_{N_1,k}(g^{-1}_{N_2,j}(t)) \tag{6}$$

Based on 5 and 7, we have

$$g_{n_1,k}(g^{-1}_{n_2,j}(t)) \geq g_{N_1,k}(g^{-1}_{N_2,j}(t)) \tag{7}$$

Therefore, when $\frac{n_1}{N_1} \geq 1 \geq \frac{n_2}{N_2}$ and $k = j$, $g_{n_1,k} \circ g^{-1}_{n_2,j}(t) - g_{N_1,k} \circ g^{-1}_{N_2,j}(t)$ is non-negative when $t \in (0,1)$.

□

Proposition 3.5. *Suppose* $\frac{n_1}{N_1} \geq \frac{n_2}{N_2}$*,* $N_1 - N_2 > n_1 - n_2$*,* $k = j$*, then* $g_{n_1,k} \circ g^{-1}_{n_2,j}(t) - g_{N_1,k} \circ g^{-1}_{N_2,j}(t)$ *is non-negative when* $t \in (0,1)$*.*

Supposed

$$f(x) = g_{n_1,k} \circ g^{-1}_{n_2,j}(x) - g_{N_1,k} \circ g^{-1}_{N_2,j}(x)$$

$$\begin{aligned} f^{'}(x) &= g^{'}_{n_1,k}(g^{-1}_{n_2,j}(x)) * \frac{1}{g^{'}_{n_2,j}(g^{-1}_{n_2,j}(x))} - g^{'}_{N_1,k}(g^{-1}_{N_2,j}(x)) * \frac{1}{g^{'}_{N_2,j}(g^{-1}_{N_2,j}(x))} \\ &= \frac{B(n_2-j, j+1)}{B(n_1-k, k+1)} g^{-1}_{n_2,j}(x)^{k-j}(1-g^{-1}_{n_2,j}(x))^{n_1-n_2} - \frac{B(N_2-j, j+1)}{B(N_1-k, k+1)} g^{-1}_{N_2,j}(x)^{k-j}(1-g^{-1}_{N_2,j}(x))^{N_1-N_2} \\ &= \frac{B(n_2-j, j+1)}{B(n_1-k, k+1)}(1-g^{-1}_{n_2,j}(x))^{n_1-n_2} - \frac{B(N_2-j, j+1)}{B(N_1-k, k+1)}(1-g^{-1}_{N_2,j}(x))^{N_1-N_2} \end{aligned}$$

When $x = 0$,

$$\begin{aligned} f^{`}(x) &= \frac{B(n_2 - j, j+1)}{B(n_1 - k, k+1)} - \frac{B(N_2 - j, j+1)}{B(N_1 - k, k+1)} \\ &= \frac{(n_2 - j - 1)!(j)!/n_2!}{(n_1 - k - 1)!(k)!/n_1!} - \frac{(N_2 - j - 1)!(j)!/N_2!}{(N_1 - k - 1)!(k)!/N_1!} \\ &= \frac{(n_2 - j)(n_2 - j + 1)\dots(n_2)}{(n_1 - j)(n_1 - j + 1)\dots(n_1)} - \frac{(N_2 - j)(N_2 - j + 1)\dots(N_2)}{(N_1 - j)(N_1 - j + 1)\dots(N_1)} \end{aligned}$$

Supposed $h(x) = \frac{n_1 - x}{n_2 - x} - \frac{N_1 - x}{N_2 - x}$, then

$$h(x) = \frac{(n_1 - x)(N_2 - x) - (n_2 - x)(N_1 - x)}{(n_2 - x)(N_2 - x)}$$

We define $l(x) = (n_1 - x)(N_2 - x) - (n_2 - x)(N_1 - x)$, it is easy to prove $l^{`}(x) > 0$ when $N_1 - N_2 > n_1 - n_2$.

Then $l(x) > l(0)$ for all $0 < x < min(n_1, n_2, N_1, N_2)$, which is $l(x) > n_1N_2 - n_2N_1$ for all $0 < x < min(n_1, n_2, N_1, N_2)$

Now we have

$$h(x) = \frac{l(x)}{(n_2 - x)(N_2 - x)} > \frac{n_1N_2 - n_2N_1}{(n_2 - x)(N_2 - x)}$$

We know

$$\frac{n_1N_2 - n_2N_1}{(n_2 - x)(N_2 - x)} > \frac{n_1N_2 - n_2N_1}{n_2N_2} = \frac{n_1}{n_2} - \frac{N_1}{N_2} > 0$$

for $0 < x < min(n_1, n_2, N_1, N_2)$.

Now we prove for all $0 < x < min(n_1, n_2, N_1, N_2)$, $h(x) > 0$.

Finally, we show

$$\begin{aligned} f^{`}(x) &= \frac{(n_2 - j)(n_2 - j + 1)\dots(n_2)}{(n_1 - j)(n_1 - j + 1)\dots(n_1)} - \frac{(N_2 - j)(N_2 - j + 1)\dots(N_2)}{(N_1 - j)(N_1 - j + 1)\dots(N_1)} \\ &= \frac{n_2 - j}{n_1 - j} \times \frac{n_2 - j + 1}{n_1 - j + 1} \times \dots \times \frac{n_2}{n_1} - \frac{N_2 - j}{N_1 - j} \times \frac{N_2 - j + 1}{N_1 - j + 1} \times \dots \times \frac{N_2}{N_1} > 0 \end{aligned}$$

When x = 1, then $f^{`}(x) = 0$. And we know when x = 0, $f(x) = 0$, and when $x = 1$, $f(x) = 0$. Therefore, $f(x) \geq 0$ for all $x \in (0, 1)$.

4 Paradox Under Different Distribution

Definition 4.1. *We will say that Y_2 is first order ε-weakly stochastically dominant over Y_1 if, for $\varepsilon > 0$, $\mathrm{P}(Y_2 \leq y) \leq \mathrm{P}(Y_1 \leq y) \leq \mathrm{P}(Y_2 \leq y) + \varepsilon$ for all y and $\mathrm{P}(Y_2 \leq y) < \mathrm{P}(Y_1 \leq y) \leq \mathrm{P}(Y_2 \leq y) + \varepsilon$ for some y.*

Proposition 4.2. *Suppose that the system inclusion mechanism has the property that* $g(X_{1,(r_{1,k})}) = 1$ *and* $g(X_{2,(r_{2,j})}) = 1$*, and that the Full House Model is of the form*

$$\begin{aligned} Y_{i,j} &\sim F_{Y_i}(y), \qquad i = 1, 2, \qquad j = 1, \ldots, n_i, \\ X_{i,j} &\sim F_X(x), \qquad i = 1, 2, \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $F_{Y_i}^{-1}(y)$ *and* $F_X^{-1}(x)$ *exist. Further suppose that* Y_2 *is first order* ε*-weakly stochastically dominant over* Y_1 *for some* $\varepsilon > 0$*. Then,*

$$\begin{aligned} \frac{n_2}{n_1 + n_2} - \frac{N_2}{N_1 + N_2} &\leq \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &\leq n_2 \sum_{i=0}^{n_1} \frac{\binom{n_1}{i}\varepsilon^i}{n_1 + n_2 - i} - \frac{N_2}{N_1 + N_2} \end{aligned} \tag{8}$$

Proof. First, observe that the assumption of first order ε-weakly stochastically dominant gives

$$\left(F_{Y_{2,(n_2)}}(y)\right)^{n_1/n_2} \leq \left(F_{Y_1}(y)\right)^{n_1} \leq \left(F_{Y_2}(y) + \varepsilon\right)^{n_1} = \left(\left(F_{Y_{2,(n_2)}}(y)\right)^{1/n_2} + \varepsilon\right)^{n_1},$$

and this implies that

$$F_X^{-1}\left(F_X^{\frac{N_2}{N_1}\frac{n_1}{n_2}}(X_{2,(N_2)})\right) \leq h_{1,2}(X_{2,(N_2)}) \leq F_{X_{1,(N_1)}}^{-1}\left(\left(\left(F_{X_{2,(N_2)}}(X_{2,(N_2)})\right)^{1/n_2} + \varepsilon\right)^{n_1}\right).$$

In combination with Corollary 3.2.1 and the proof of Corollary 3.2.1, the first inequality yields

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \geq \frac{n_2}{n_1 + n_2} - \frac{N_2}{N_1 + N_2}.$$

This verifies the first inequality of (8). We now verify the second inequality of (8). Now,

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &\quad \leq \int_{-\infty}^{\infty} \left[\int_x^{F_{X_{1,(N_1)}}^{-1}\left(\left(\left(F_{X_{2,(N_2)}}(x)\right)^{1/n_2} + \varepsilon\right)^{n_1}\right)} f_{X_{1,(N_1)}}(y)dy\right] f_{X_{2,(N_2)}}(x)dx \\ &\quad = \int_{-\infty}^{\infty} \left[\left(\left(F_X(x)\right)^{N_2/n_2} + \varepsilon\right)^{n_1} - F_X(x)^{N_1}\right] N_2 F_X(x)^{N_2 - 1} f_X(x)dx \\ &\quad = \int_{-\infty}^{\infty} \left(\left(F_X(x)\right)^{N_2/n_2} + \varepsilon\right)^{n_1} N_2 F_X(x)^{N_2 - 1} f_X(x)dx - \frac{N_2}{N_1 + N_2} \\ &\quad = \int_0^1 \left(t^{1/n_2} + \varepsilon\right)^{n_1} dt - \frac{N_2}{N_1 + N_2} \\ &\quad = n_2 \sum_{i=0}^{n_1} \frac{\binom{n_1}{i}\varepsilon^i}{n_1 + n_2 - i} - \frac{N_2}{N_1 + N_2} \end{aligned}$$

□

Example: Uniform distributions. In this example we demonstrate a case where Prop 4.2 applies, but we can explicitly calculate the paradox probability. Suppose that the system inclusion mechanism has the property that $g(X_{i,(N_i)}, \mathbf{X}_{i,-(N_i)}) = 1$, and we will consider the following Full House Model for two systems,

$$\begin{aligned} Y_{1,j} &\sim U(0,1), \qquad j = 1, \ldots, n_1, \\ Y_{2,j} &\sim U(0, 1+\delta), \qquad j = 1, \ldots, n_2, \\ X_{i,j} &\sim F_X(x), \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $\delta > 0$. We can pick $\varepsilon = \delta/(1+\delta)$ so that Y_2 is first order ε-weakly stochastically dominant over Y_1 with this choice of ε, and the conditions of Prop 4.2 are satisfied. However, our setup allows for direct calculation of the paradox probability. From Lemma 3.1 and a bit of manipulation, we have that

Theorem 4.3.

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \begin{cases} \frac{n_2}{n_1+n_2}(1+\delta)^{n_1} - \frac{N_2}{N_1+N_2}, & \text{if } \frac{n_1}{N_1} \geq \frac{n_2}{N_2} \\ \frac{N_1}{N_1+N_2} - \frac{n_1}{n_1+n_2}(\frac{1}{(1+\delta)})^{n_2} - & \\ \frac{n_2}{n_1+n_2}(1+\delta)^{n_1} d^{\frac{(n_1+n_2)N_2}{n_2}} + \frac{N_2}{N_1+N_2} d^{(N_1+N_2)}, & \text{otherwise} \end{cases}.$$

Proof.

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &= \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}, Y_{2,(n_2)} \in (0,1)\right) \\ &+ \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}, Y_{2,(n_2)} \in (1,1+\delta)\right) \\ &= \mathrm{P}\left(X_{2,(N_2)} < X_{1,(N_1)} < f_{1,2}(X_{2,(N_2)}), F_X(X_2)^{\frac{N_2}{n_2}}(1+\delta) \in (0,1)\right) \\ &+ \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, F_X(X_2)^{\frac{N_2}{n_2}}(1+\delta) \in (1,1+\delta)\right) \\ &= \int_{-\infty}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} \int_{x}^{F_X^{-1}((F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{\frac{n_1}{N_1}})} d(F_X(y)^{N_1})\, d(F_X(x)^{N_2}) \\ &+ \int_{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})}^{+\infty} \int_{x}^{+\infty} d(F_X(y)^{N_1})\, d(F_X(x)^{N_2}) \\ &= \int_{-\infty}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} \int_{x}^{F_X^{-1}((F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{\frac{n_1}{N_1}})} d(F_X(y)^{N_1})\, d(F_X(x)^{N_2}) \\ &+ (1 - (\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1 - (\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \end{aligned}$$

The support for the outer integration above is $\{F_X^{-1}((F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{\frac{n_1}{N_1}}) > x\} \cup \{x < F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})\}$.

Therefore, when $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$,

$$
\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \int_{-\infty}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} \int_{x}^{F_X^{-1}((F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{\frac{n_1}{N_1}})} d(F_X(y)^{N_1})\ d(F_X(x)^{N_2}) \\
&+ (1-(\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1-(\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \\
&= \int_{-\infty}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} (F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{n_1} - F_X(x)^{N_1}\ d(F_X(x)^{N_2}) \\
&+ (1-(\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1-(\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \\
&= \int_{0}^{(\frac{1}{1+\delta})^{n_2}} (1+\delta)^{n_1} t^{\frac{n_1}{n_2}} - t^{\frac{N_1}{N_2}}\ d(F_X(x)^{N_2}) \\
&+ (1-(\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1-(\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \\
&= \frac{N_1}{N_1+N_2} - (\frac{1}{1+\delta})^{n_2}\frac{n_1}{n_1+n_2}
\end{aligned}
$$

Therefore, when $\frac{n_1}{N_1} \geq \frac{n_2}{N_2}$, and let $d = (1+\delta)^{\frac{1}{\frac{N_1}{n_1}-\frac{N_2}{n_2}}}$

$$
\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \int_{F_X^{-1}(d)}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} \int_{x}^{F_X^{-1}((F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{\frac{n_1}{N_1}})} d(F_X(y)^{N_1})\ d(F_X(x)^{N_2}) \\
&+ (1-(\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1-(\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \\
&= \int_{F_X^{-1}(d)}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} (F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{n_1} - F_X(x)^{N_1}\ d(F_X(x)^{N_2}) \\
&+ (1-(\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1-(\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \\
&= \int_{d^{N_2}}^{(\frac{1}{1+\delta})^{n_2}} (1+\delta)^{n_1} t^{\frac{n_1}{n_2}} - t^{\frac{N_1}{N_2}}\ d(F_X(x)^{N_2}) \\
&+ (1-(\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1-(\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \\
&= \frac{N_1}{N_1+N_2} - \frac{n_1}{n_1+n_2}(\frac{1}{(1+\delta)})^{n_2} - \frac{n_2}{n_1+n_2}(1+\delta)^{n_1} d^{\frac{(n_1+n_2)N_2}{n_2}} + \frac{N_2}{N_1+N_2} d^{(N_1+N_2)}
\end{aligned}
$$

□

Here we see that the probability of paradox is 1 for all $\delta \geq [2(n_1/n_2+1)]^{1/n_1}-1$. When $n_1 \gg n_2$, then $[2(n_1/n_2+1)]^{1/n_1}-1 \approx n_1^{n_1}-1$ and we see that the probability of paradox is 1 for extremely small values of δ.

Theorem 4.4. *Assume $g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1}(t) - g_{N_1,k} \circ g_{N_2,j}^{-1}(t)$ is non-negative when $t \in (0,1)$. Then*

$$\begin{aligned}\mathrm{P}\left(X_{1,(N_1-k)} > X_{2,(N_2-j)}, Y_{1,(n_1-k)} < Y_{2,(n_2-j)}\right) &\leq (1+\delta)^p \sum_{p=n_1-k}^{n_1} \left(\begin{array}{c} n_1 \\ p \end{array} \right) \frac{B(n_2-j+p, n_1-p+j+1)}{B(n_2-j, j+1)} \\ &- \sum_{p=N_1-k}^{N_1} \left(\begin{array}{c} N_1 \\ p \end{array} \right) \frac{B(N_2-j+p, N_1-p+j+1)}{B(N_2-j, j+1)}\end{aligned}$$

Proof.

$$\begin{aligned}&\mathrm{P}\left(X_{1,(N_1-k)} > X_{2,(N_2-j)}, Y_{1,(n_1-k)} < Y_{2,(n_2-j)}\right) = \mathrm{P}\left(X_{2,(N_2-j)} < X_{1,(N_1-k)} < h_{1,2}\left(X_{2,(N_2-j)}\right)\right) \\
&= \int_{-\infty}^{+\infty} \left[\int_x^{F_X^{-1} \circ g_{N_1,k}^{-1} \circ g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1} \circ g_{N_2,j} \circ F_X(x)} f_{X_{1,(N_1-k)}}(y)\ dy \right] f_{X_{2,(N_2-j)}}(x)\ dx \\
&= \int_{-\infty}^{+\infty} \left[\int_x^{F_X^{-1} \circ g_{N_1,k}^{-1} \circ g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1} \circ g_{N_2,j} \circ F_X(x)} d\ g_{N_1,k}(F_X(y)) \right] f_{X_{2,(N_2-j)}}(x)\ dx \\
&= \int_{-\infty}^{+\infty} \left[g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1} \circ g_{N_2,j} \circ F_X(x) - g_{N_1,k} \circ F_X(x) \right]^+ f_{X_{2,(N_2-j)}}(x)\ dx \\
&= \int_{-\infty}^{+\infty} \left[g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1} \circ g_{N_2,j} \circ F_X(x) - g_{N_1,k} \circ F_X(x) \right]^+ d\ g_{N_2,j}(F_X(x)) \\
&= \int_0^1 \left[g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1}(t) - g_{N_1,k} \circ g_{N_2,j}^{-1}(t) \right]^+ dt \\
&= \int_0^1 g_{n_1,k}((1+\delta)x)\ dg_{n_2,j}(x) - \int_0^1 g_{N_1,k}(x)\ dg_{N_2,j}(x) \\
&= \int_0^1 I_{(1+\delta)x}(n_1-k, k+1) \frac{x^{n_2-j-1}(1-x)^j}{B(n_2-j, j+1)}\ dx - \int_0^1 I_x(N_1-k, k+1) \frac{x^{N_2-j-1}(1-x)^j}{B(N_2-j, j+1)}\ dx \\
&= \int_0^1 \sum_{p=n_1-k}^{n_1} \left(\begin{array}{c} n_1 \\ p \end{array} \right) ((1+\delta)x)^p (1-(1+\delta)x)^{n_1-p} \frac{x^{n_2-j-1}(1-x)^j}{B(n_2-j, j+1)}\ dx \\
&- \int_0^1 \sum_{p=N_1-k}^{N_1} \left(\begin{array}{c} N_1 \\ p \end{array} \right) x^p (1-x)^{N_1-p} \frac{x^{N_2-j-1}(1-x)^j}{B(N_2-j, j+1)}\ dx \\
&\leq (1+\delta)^p \int_0^1 \sum_{p=n_1-k}^{n_1} \left(\begin{array}{c} n_1 \\ p \end{array} \right) x^p (1-x)^{n_1-p} \frac{x^{n_2-j-1}(1-x)^j}{B(n_2-j, j+1)}\ dx \\
&- \int_0^1 \sum_{p=N_1-k}^{N_1} \left(\begin{array}{c} N_1 \\ p \end{array} \right) x^p (1-x)^{N_1-p} \frac{x^{N_2-j-1}(1-x)^j}{B(N_2-j, j+1)}\ dx\end{aligned}$$

$$= (1+\delta)^p \sum_{p=n_1-k}^{n_1} \left(\begin{array}{c} n_1 \\ p \end{array} \right) \frac{B(n_2-j+p, n_1-p+j+1)}{B(n_2-j, j+1)} - \sum_{p=N_1-k}^{N_1} \left(\begin{array}{c} N_1 \\ p \end{array} \right) \frac{B(N_2-j+p, N_1-p+j+1)}{B(N_2-j, j+1)}$$

Example: Normal distributions

In this example, we demonstrate a case where Prop 4.2 applies. Suppose that the system inclusion mechanism has the property that $g(X_{i,(N_i)}, \mathbf{X}_{i,-(N_i)}) = 1$, and we will consider the following Full House Model for two systems,

$$\begin{aligned} Y_{1,j} &\sim N(\mu_1, \sigma_1^2), \qquad j = 1, \ldots, n_1, \\ Y_{2,j} &\sim N(\mu_2, \sigma_2^2), \qquad j = 1, \ldots, n_2, \\ X_{i,j} &\sim F_X(x), \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $\mu_1 \leq \mu_2, \sigma_1 > 0$ and $\sigma_2 > 0$.

Theorem 4.5. *Assume* $\Phi(\frac{\Phi^{-1}(t^{\frac{N_2}{n_2}})\sigma_2 + \mu_2 - \mu_1}{\sigma_1})^{\frac{n_1}{N_1}} > t$ *for all* $t \in (0,1)$,

When $\sigma_1 = \sigma_2 = \sigma$,

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &= n_2 F_{\mathbf{W}}(\mathbf{w} = (\frac{\mu_2 - \mu_1}{\sigma}, \ldots, \frac{\mu_2 - \mu_1}{\sigma}, 0, \ldots, 0)) - \frac{N_2}{N_1 + N_2}. \end{aligned}$$

$\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ *with* (n_1+n_2-1)*-dimensional mean vector*

$$\boldsymbol{\mu} = (0, \ldots, 0)$$

and $(n_1+n_2-1) \times (n_1+n_2-1)$ *covariance matrix*

$$\boldsymbol{\Sigma} = \begin{bmatrix} 2 & 1 & \ldots & 1 \\ 1 & 2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \ldots & 1 & 2 \end{bmatrix}$$

The first n_1 *elements of* $\mathbf{w}$ *is* $\frac{\mu_2 - \mu_1}{\sigma}$ *and the last* $n_2 - 1$ *elements are 0.*

When $\sigma_1 \neq \sigma_2$,

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &= n_2 F_{\mathbf{W}}(\mathbf{w} = (\frac{\mu_2 - \mu_1}{\sigma_1}, \ldots, \frac{\mu_2 - \mu_1}{\sigma_1}, 0, \ldots, 0)) - \frac{N_2}{N_1 + N_2}. \end{aligned}$$

$\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ *with* (n_1+n_2-1)*-dimensional mean vector*

$$\boldsymbol{\mu} = (0, \ldots, 0)$$

and $(n_1+n_2-1)\times(n_1+n_2-1)$ *covariance matrix*

$$\mathbf{\Sigma} = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

$$\Sigma_{11} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}_{n_1\times n_1}, \Sigma_{12}^T = \Sigma_{21} = \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix}_{(n_2-1)\times n_1}$$

$$\Sigma_{22} = \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}_{(n_2-1)\times(n_2-1)}$$

The first n_1 *elements of* $\mathbf{w}$ *is* $\frac{\mu_2-\mu_1}{\sigma_1}$ *and the last* n_2-1 *elements are 0.*

Proof. Supposed $\sigma_1=\sigma_2=\sigma$, it is obvious to see $\mathrm{P}(Y_2\le y)\le \mathrm{P}(Y_1\le y)$ and we can also find the ε such that $\mathrm{P}(Y_1\le y)\le \mathrm{P}(Y_2\le y)+\varepsilon$

$$\begin{aligned} F_{Y_1}(y)-F_{Y_2}(y) &= \int_{-\infty}^{y}\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{(x-\mu_1)^2}{2\sigma^2}}dx - \int_{-\infty}^{y}\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{(x-\mu_2)^2}{2\sigma^2}}dx \\ &= \int_{y-\frac{\mu_2-\mu_1}{2}}^{y+\frac{\mu_2-\mu_1}{2}}\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{(x-\frac{\mu_2+\mu_1}{2})^2}{2\sigma^2}}dx \end{aligned}$$

Since $\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{(x-\frac{\mu_2+\mu_1}{2})^2}{2\sigma^2}}$ is the density function for $N(\frac{\mu_2+\mu_1}{2},\sigma^2)$, then the maximum value of the difference between these two cumulative distribution function of the normal distribution is achieved when $y=\frac{\mu_2+\mu_1}{2}$.

We can pick $\varepsilon=\int_{\mu_1}^{\mu_2}\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{(x-\frac{\mu_2+\mu_1}{2})^2}{2\sigma^2}}dx$ so that Y_2 is first order ε-weakly stochastically dominant over Y_1 with this choice of ε, and the conditions of Prop 4.2 are satisfied.

From Lemma 3.1 and a bit of manipulation, we have that

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)}>X_{2,(N_2)}, Y_{1,(n_1)}<Y_{2,(n_2)}\right) = \mathrm{P}\left(X_{2,(N_2)}<X_{1,(N_1)}<h_{1,2}\left(X_{2,(N_2)}\right)\right) \\ &= \int_{-\infty}^{+\infty}\int_{x}^{h_{1,2}(x)} f_{X_{1,(N_1)}}(y)\ dy\ f_{X_{2,(N_2)}}(x)\ dx \\ &= \int_{-\infty}^{+\infty}\int_{x}^{h_{1,2}(x)} N_1(F_X(y))^{N_1-1}\ dy\ N_2F_X(x)^{N_2-1}\ dx \\ &= \int_{-\infty}^{+\infty}\int_{x}^{F_X^{-1}((\Phi(\Phi^{-1}(F_X^{\frac{N_2}{n_2}})+\frac{\mu_2-\mu_1}{\sigma}))^{\frac{n_1}{N_1}})(x)} N_1F_X(y)^{N_1-1}\ dy\ N_2F_X(x)^{N_2-1}\ dx \end{aligned}$$

$$= \int_{-\infty}^{+\infty} ((\Phi(\Phi^{-1}(F_X(x)^{\frac{N_2}{n_2}}) + \frac{\mu_2 - \mu_1}{\sigma}))^{n_1} - F_X(x)^{N_1})\ N_2 F_X(x)^{N_2-1}\ dx$$

$$= \int_0^1 (\Phi(\Phi^{-1}(t^{\frac{1}{n_2}}) + \frac{\mu_2 - \mu_1}{\sigma}))^{n_1} - t^{\frac{N_1}{N_2}})\ dt$$

$$= \int_0^1 (\Phi(\Phi^{-1}(t^{\frac{1}{n_2}}) + \frac{\mu_2 - \mu_1}{\sigma}))^{n_1}\ dt - \frac{N_2}{N_1 + N_2}$$

$$= \int_{-\infty}^{+\infty} \Phi(x + \frac{\mu_2 - \mu_1}{\sigma})^{n_1} (n_2 \Phi(x)^{n_2-1}\phi(x))\ dx - \frac{N_2}{N_1 + N_2}$$

$$= n_2 \int_{-\infty}^{+\infty} \mathbb{P}(Z_{1,1} - X \leq \frac{\mu_2 - \mu_1}{\sigma}, \ldots, Z_{1,n_1} - X \leq \frac{\mu_2 - \mu_1}{\sigma},$$

$$Z_{2,1} - X \leq 0, \ldots, Z_{2,n_2-1} - X \leq 0 | X = x)\phi(x)\ dx - \frac{N_2}{N_1 + N_2}$$

$$= n_2 \mathbb{P}(Z_{1,1} - X \leq \frac{\mu_2 - \mu_1}{\sigma}, \ldots, Z_{1,n_1} - X \leq \frac{\mu_2 - \mu_1}{\sigma}, Z_{2,1} - X \leq 0, \ldots, Z_{2,n_2-1} - X \leq 0) - \frac{N_2}{N_1 + N_2}$$

$$= n_2 F_{\mathbf{W}}(\mathbf{w} = (\frac{\mu_2 - \mu_1}{\sigma}, \ldots, \frac{\mu_2 - \mu_1}{\sigma}, 0, \ldots, 0)) - \frac{N_2}{N_1 + N_2}$$

where

$$\begin{aligned} h_{1,2}(x) &= (g_1 \circ f_2)(x) \\ &= F_X^{-1}((F_{Y_2}(F_{Y_1}^{-1}(F_X^{\frac{N_2}{n_2}})))^{\frac{n_1}{N_1}})(x) \\ &= F_X^{-1}((\Phi(\Phi^{-1}(F_X^{\frac{N_2}{n_2}}) - \frac{\mu}{\sigma}))^{\frac{n_1}{N_1}})(x) \end{aligned}$$

$\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $(n_1 + n_2 - 1)$-dimensional mean vector

$$\boldsymbol{\mu} = (0, \ldots, 0)$$

and $(n_1 + n_2 - 1) \times (n_1 + n_2 - 1)$ covariance matrix

$$\boldsymbol{\Sigma} = \begin{bmatrix} 2 & 1 & \ldots & 1 \\ 1 & 2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \ldots & 1 & 2 \end{bmatrix}$$

The first n_1 elements of $\mathbf{w}$ is $\frac{\mu_2 - \mu_1}{\sigma}$ and the last $n_2 - 1$ elements are 0.

Then when $\sigma_1 \neq \sigma_2$, we do not have a closed form for ϵ but we can calculate the probability of the paradox.

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(X_{2,(N_2)} < X_{1,(N_1)} < h_{1,2}\left(X_{2,(N_2)}\right)\right)$$

$$= \int_{-\infty}^{+\infty} \int_x^{h_{1,2}(x)} f_{X_{1,(N_1)}}(y)\ dy\ f_{X_{2,(N_2)}}(x)\ dx$$

$$= \int_{-\infty}^{+\infty} \int_{x}^{h_{1,2}(x)} N_1(F_X(y))^{N_1-1}\ dy\ N_2 F_X(x)^{N_2-1}\ dx$$

$$= \int_{-\infty}^{+\infty} \int_{x}^{F_X^{-1}(\Phi(\frac{\Phi^{-1}(F_X^{\frac{N_2}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1})^{\frac{n_1}{N_1}})(x)} N_1 F_X(y)^{N_1-1}\ dy\ N_2 F_X(x)^{N_2-1}\ dx$$

$$= \int_{-\infty}^{+\infty} ((\Phi(\frac{(\Phi^{-1}(F_X(x)^{\frac{N_2}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{n_1} - F_X(x)^{N_1})\ N_2 F_X(x)^{N_2-1}\ dx$$

$$= \int_{0}^{1} (\Phi(\frac{\Phi^{-1}(t^{\frac{1}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{n_1} - t^{\frac{N_1}{N_2}})\ dt$$

$$= \int_{0}^{1} (\Phi(\frac{\Phi^{-1}(t^{\frac{1}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{n_1}\ dt - \frac{N_2}{N_1+N_2}$$

$$= \int_{-\infty}^{+\infty} \Phi(\frac{\sigma_2 x+\mu_2-\mu_1}{\sigma_1})^{n_1}(n_2\Phi(x)^{n_2-1}\phi(x))\ dx - \frac{N_2}{N_1+N_2}$$

$$= n_2 \int_{-\infty}^{+\infty} \mathbb{P}(Z_{1,1} - \frac{\sigma_2}{\sigma_1}X \leq \frac{\mu_2-\mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1}X \leq -\frac{\mu_2-\mu_1}{\sigma_1},$$

$$Z_{2,1} - X \leq 0, \ldots, Z_{2,n_2-1} - X \leq 0 | X = x)\phi(x)\ dx - \frac{N_2}{N_1+N_2}$$

$$= n_2 \mathbb{P}(Z_{1,1} - \frac{\sigma_2}{\sigma_1}X \leq \frac{\mu_2-\mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1}X \leq -\frac{\mu_2-\mu_1}{\sigma_1}, Z_{2,1} - X \leq 0, \ldots, Z_{2,n_2-1} - X \leq 0)$$

$$- \frac{N_2}{N_1+N_2}$$

$$= n_2 F_{\mathbf{W}}(\mathbf{w} = (\frac{\mu_2-\mu_1}{\sigma_1}, \ldots, \frac{\mu_2-\mu_1}{\sigma_1}, 0, \ldots, 0)) - \frac{N_2}{N_1+N_2}$$

where

$$\begin{aligned} h_{1,2}(x) &= (g_1 \circ f_2)(x) \\ &= F_X^{-1}((F_{Y_2}(F_{Y_1}^{-1}(F_X^{\frac{N_2}{n_2}})))^{\frac{n_1}{N_1}})(x) \\ &= F_X^{-1}((\Phi(\frac{\Phi^{-1}(F_X^{\frac{N_2}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{\frac{n_1}{N_1}})(x) \end{aligned}$$

$$\Phi^{-1}(F_X(x)^{\frac{N_1}{n_1}})\sigma_1 + \mu_1 < \Phi^{-1}(F_X(x)^{\frac{N_2}{n_2}})\sigma_2 + \mu_2$$

$\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with (n_1+n_2-1)-dimensional mean vector

$$\boldsymbol{\mu} = (0, \ldots, 0)$$

and $(n_1+n_2-1) \times (n_1+n_2-1)$ covariance matrix

$$\boldsymbol{\Sigma} = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

$$\Sigma_{11} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}_{n_1 \times n_1}, \Sigma_{12}^T = \Sigma_{21} = \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix}_{(n_2-1)\times n_1}$$

$$\Sigma_{22} = \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}_{(n_2-1)\times(n_2-1)}$$

The first n_1 elements of $\mathbf{w}$ is $\frac{\mu_2-\mu_1}{\sigma_1}$ and the last n_2-1 elements are 0.

□

5 Simulation

In this section, We verify Theorem 3.2, Theorem 4.3, and 4.5 in the numerical simulation. We first randomly generate two groups of observation: Y_1 and Y_2 from the distributions below, and the number of each group is given by n_1 and n_2. Based on our setting, we assume the latent aptitudes follow the Pareto distribution with $\alpha = 1.16$, and $r(N_i, n_i, t) = N_i - n_i + t$. Under these assumptions, we select the top n_i talents from the talent pools X_i. Then we obtain the latent aptitudes for each group and check whether the paradox happens. We repeat this process 1000 times and record the frequency of the paradox. 100 Monte Carlo iterations of our simulation are performed and we compute the average of the frequency.

Setting	Distribution for Y_1	Distribution for Y_2	Prob from Theorem	Prob from simulation
$N_1 = 1e3, N_2 = 1e3, k = n_1 - 1, j = n_2 - 1$	$N(0,1), n_1 = 40$	$N(0,1), n_2 = 60$	0.1495	0.1472
$N_1 = 1e3, N_2 = 1e3, k = n_1 - 1, j = n_2 - 1$	$N(0,1), n_1 = 30$	$N(0,1), n_2 = 70$	0.2866	0.2875
$N_1 = 1e3, N_2 = 1e3, k = n_1 - 1, j = n_2 - 1$	$N(0,1), n_1 = 60$	$N(0,1), n_2 = 40$	0	0
$N_1 = 1e3, N_2 = 1e3, k = n_1 - 1, j = n_2 - 1$	$N(0,1), n_1 = 70$	$N(0,1), n_2 = 30$	0	0
$N_1 = 1e5, N_2 = 1e5, k = n_1, j = n_2$	$U(0,1), n_1 = 40$	$U(0,1+\delta), n_2 = 60$	0.1233	0.1224
$N_1 = 1e5, N_2 = 1e5, k = n_1, j = n_2$	$U(0,1), n_1 = 30$	$U(0,1+\delta), n_2 = 70$	0.2203	0.2210
$N_1 = 1e5, N_2 = 1e5, k = n_1, j = n_2$	$U(0,1), n_1 = 60$	$U(0,1+\delta), n_2 = 40$	0.0022	0.0021
$N_1 = 1e5, N_2 = 1e5, k = n_1, j = n_2$	$U(0,1), n_1 = 70$	$U(0,1+\delta), n_2 = 30$	0.0008	0.0007
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,2), n_1 = 40$	$N(1,4), n_2 = 60$	0.3338	0.3356
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,2), n_1 = 30$	$N(1,4), n_2 = 70$	0.3336	0.3343
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,2), n_1 = 10$	$N(1,4), n_2 = 100$	0.3331	0.3345
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,2), n_1 = 100$	$N(1,4), n_2 = 10$	0.2378	0.2406
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,1), n_1 = 40$	$N(1,1), n_2 = 60$	0.3321	0.3326
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,1), n_1 = 30$	$N(1,1), n_2 = 70$	0.3313	0.3325
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,1), n_1 = 10$	$N(1,1), n_2 = 100$	0.3334	0.3346
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,1), n_1 = 100$	$N(1,1), n_2 = 10$	0.2593	0.2592

Table 1: Simulation result to validate the Theorem 3.2

6 Application

6.1 Gould Paradox in Major League Baseball

In this analysis, we compute the probability of the paradox using the results from the Normal distribution example above and illustrate it in the application of Major League Baseball (MLB) since it is widely acknowledged that the batting average follows Normal distribution.

Specifically, we compute the probability that the top batting average in 2023 is lower than the top batting average in the other season, but the players who have the highest batting average in 2023 are more talented than those who have the highest batting average in the other season.

In this analysis, we detail the considerations made for the batting average statistics that we era-adjust using the Full House Model [Eck, 2020b]. Park factor adjustment is considered in this analysis and we apply the adjusted park index from Schell [2016] to all ballparks from the 1871 season to the 2022 season. The batting average can be affected by the ballpark and we apply the park factor adjustment to this statistic, which accounts for the handedness of batters.

The Full House Model will only be applied to the statistics obtained by full-time players. We define the full-time hitter cutoff as the median PAs after screening out hitters who appeared in fewer than 75 PAs. This criterion is flexible enough to account for changing the number of games played over time as well as seasons shortened by labor strikes and pandemics.

In the 1994 and 1995[1], hitters and pitchers played fewer games per season than the regular season. To deal with extreme statistics from the small sample size, we employed a shrinkage method that adjusted the raw statistics toward a global average. Our shrinkage method follows a shrinkage of ballpark effect estimates motivated from Schell [2016]. The details can be found in Eck [2020b].

We compute the talent scores for the batting average in each season using the (1). Then the theoretical results from the Normal distribution example above can be applied to this example and compute the paradox probability of all the seasons from 1871 to 2022 compared with 2023. Fig 1 displays the tendency of the paradox probability of all the seasons from 1871 to 2022 compared with 2023.

[1]https://en.wikipedia.org/wiki/1994%E2%80%9395_Major_League_Baseball_strike

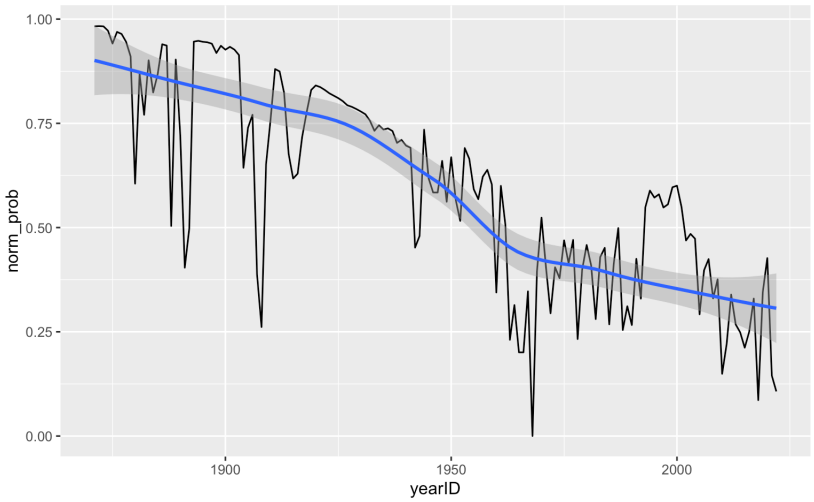


Figure 1: The paradox probability of all the seasons from 1871 to 2022 compared with 2023

When $\frac{n_1}{N_1} \geq \frac{n_2}{N_2}$ and $k = j$, then
$g_{n_1,k} \circ g^{-1}_{n_2,j}(t) - g_{N_1,k} \circ g^{-1}_{N_2,j}(t)$ is non-negative when $t \in (0,1)$, where $g_{n,i}(x) = I_x(i, n-i+1)$, $I_x(\alpha, \beta)$ is the CDF of Beta(α, β) distribution.

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